

# Predicting Traffic Accident Severity in Brazil

A Multiclass Machine Learning Approach with Imbalance Handling

Marek Martinák • D00250465 • MSc Data Analytics, DkIT • June 2026

## Background

In Brazil, traffic accidents are considered the tenth leading cause of death, claiming approximately 33,000 lives each year (Alves et al. 2021). The costs associated with traffic accidents in Brazil amount to around 220 billion reais annually, representing up to 4% of Brazil's GDP (Nunes & Silva 2021). Furthermore, studies focus mainly on developed countries. Low- and middle-income countries are thus neglected, even though they account for a significant proportion of severe traffic accidents (Alves et al. 2021). This is the problem this project aims to address.

## Research questions

- When it comes to predicting the multiclass severity of accidents using macro-F1 and per-class metrics, can tree-based classifiers (Decision Tree, Random Forest, XGBoost, LightGBM, and CatBoost) outperform a Logistic Regression baseline?
- How does handling class imbalance (using random oversampling, random undersampling, SMOTE, ADASYN, SMOTE+Tomek, and class weighting) affect the performance of these models, and how do the main methods of class imbalance handling compare?
- Which features are most important in predicting traffic accident severity?

## Hypothesis

The imbalance handling methods will improve models and their results over the base non-handling models. SHAP is going to select and identify the dominant predictors which are the biggest contributors to traffic accidents.

## Literature gap

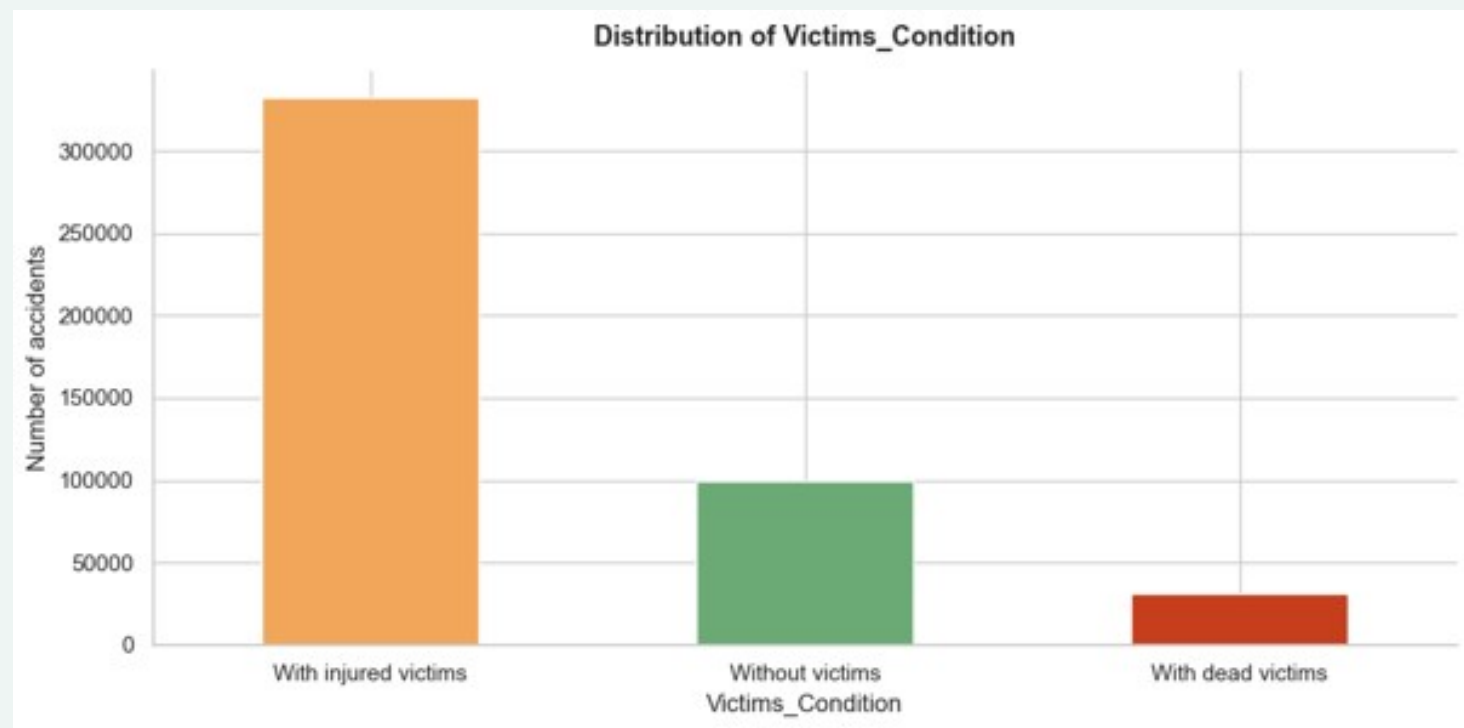
Previous studies on PRF have focused on binary classification (Scholz et al., 2024; Gabardo et al., 2026) and either ignored class imbalance, or used only a single method. None of them applied SHAP at the country level. This work addresses all three gaps: multi-class severity, a systematic comparison of imbalanced handling techniques and modern explainability.

## Dataset

Brazilian Federal Highway Police (PRF), publicly available via Kaggle.

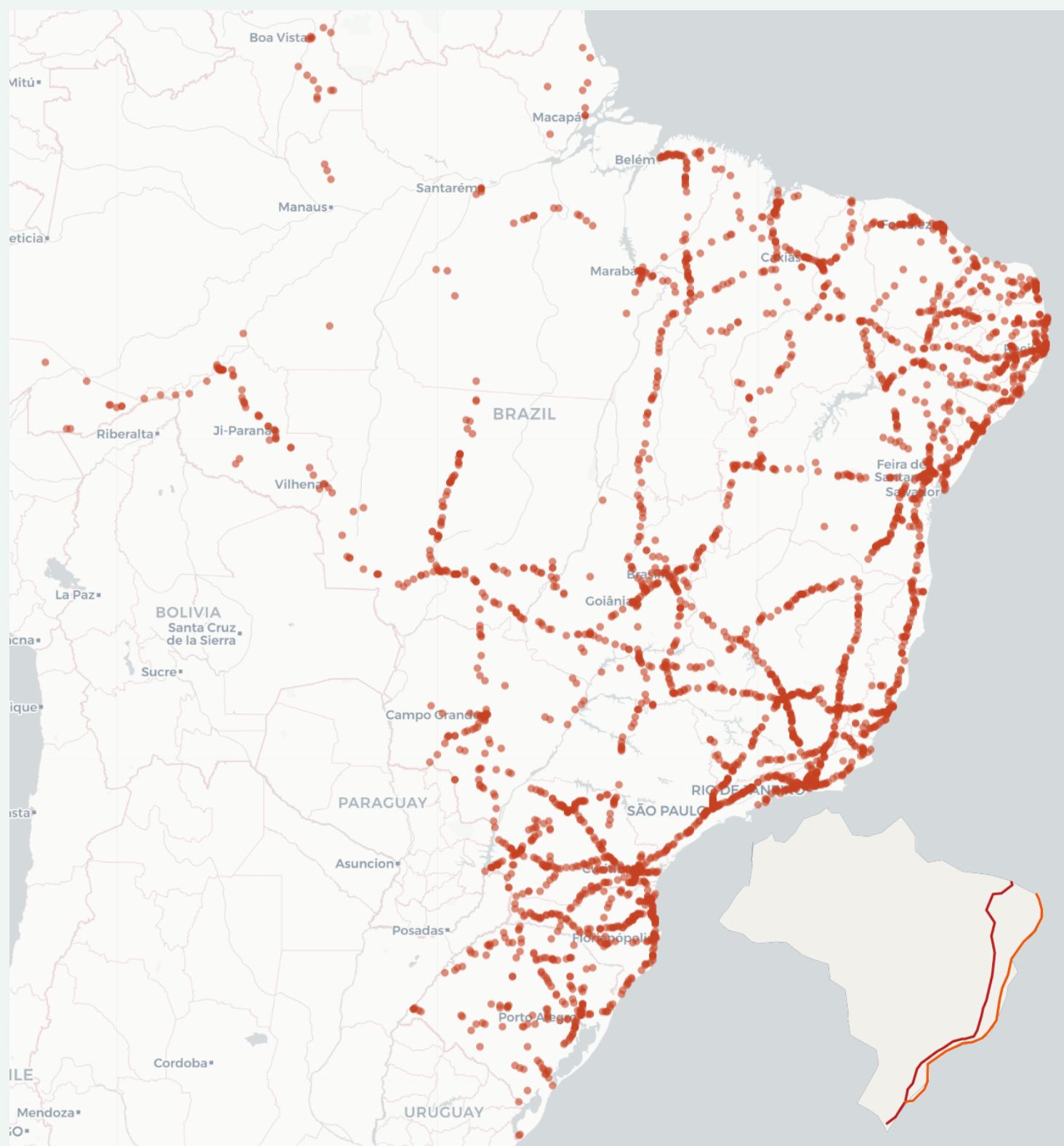
- Period:** 2017–2023
- Records:** 463,133 after cleaning
- Columns:** 27 raw → 25 engineered
- Target:** 3-class severity (With dead victims / With injured victims / Without victims)
- Fully anonymised — no personal identifiers.

## Target distribution



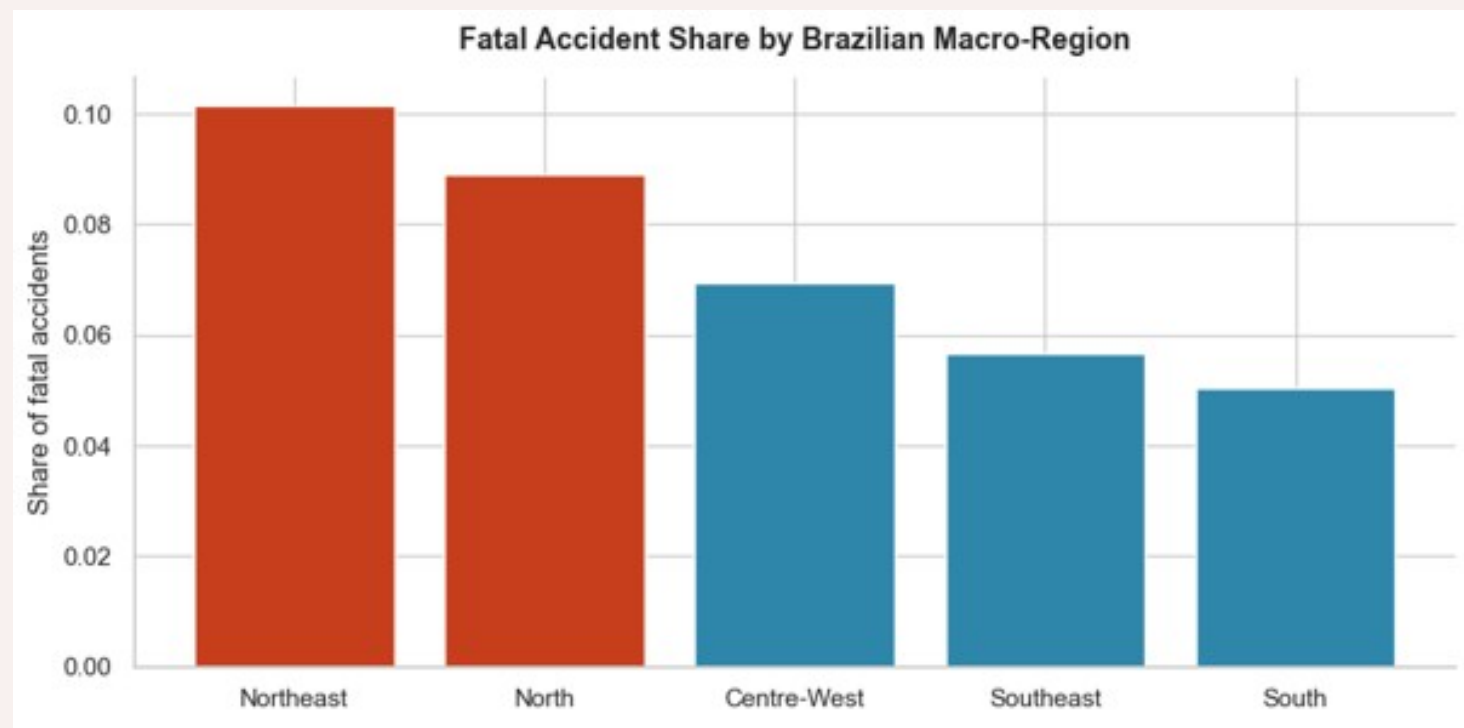
Imbalanced Target Variable: 71.7% With injured victims, 21.5% Without victims, 6.8% With dead victims. A naive classifier would hit 71.7% accuracy with zero practical value — macro-F1 is the primary metric.

## Where accidents happen



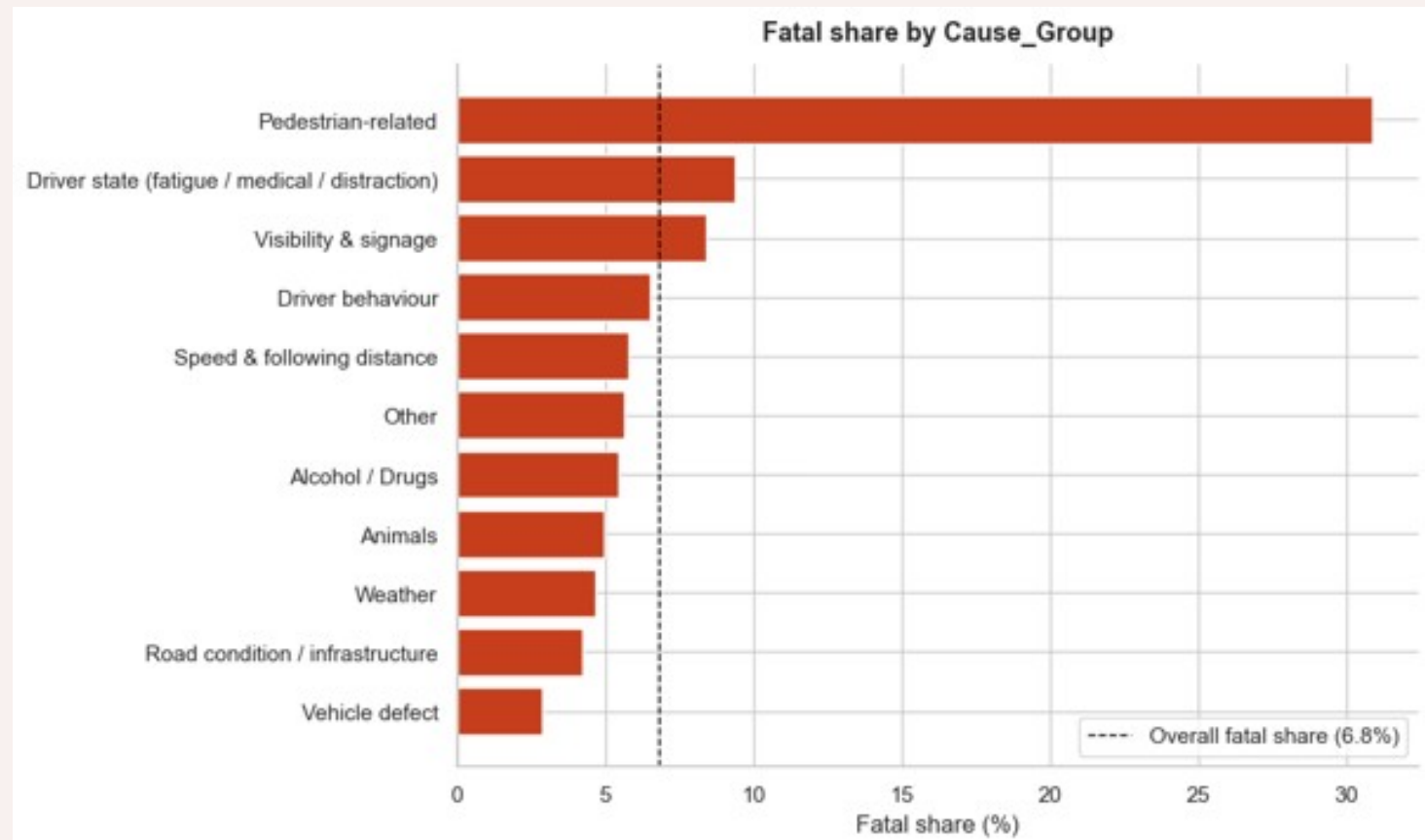
Fatal accidents concentrate on Brazil's two longest highways. BR-116 and BR-101 — account for 7,917 fatal incidents combined.

## Regional disparity



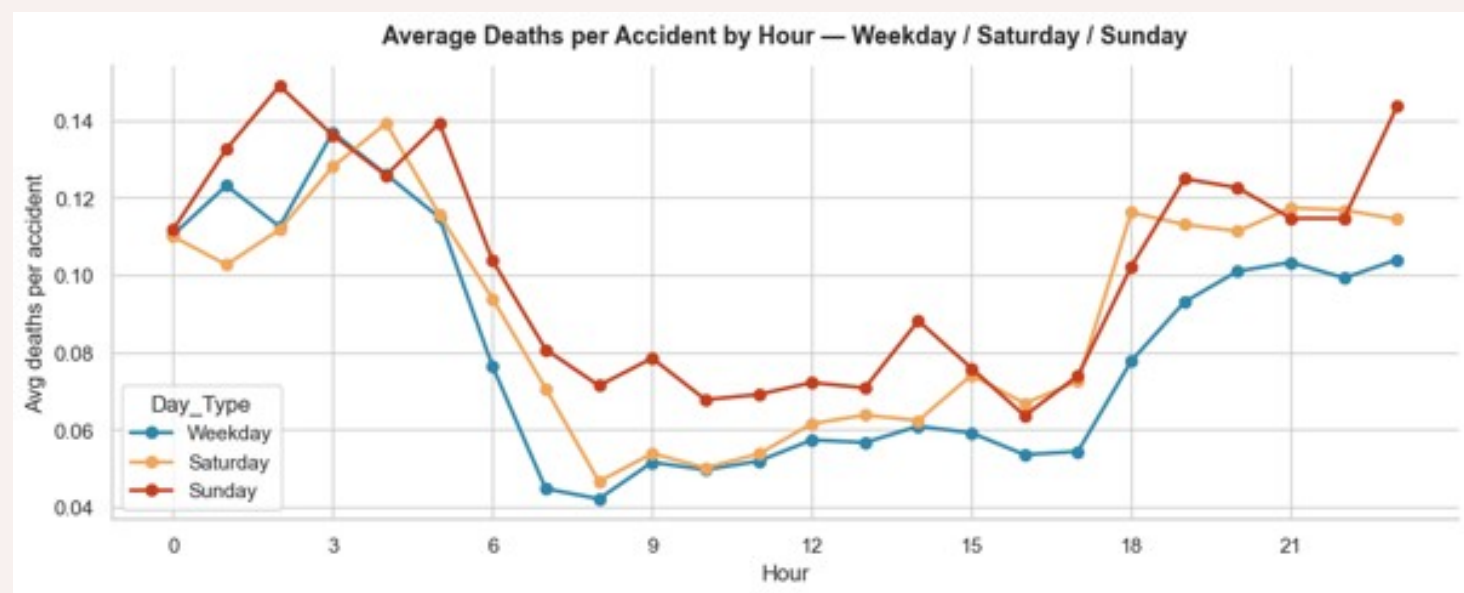
Comparing data across the five macro-regions in Brazil revealed the proportion of deaths. The Northeast carries a 10.2% fatal share, roughly twice that of the South (5.1%). This is also confirmed by the context mentioned in the Background, which states that a significant proportion of fatal traffic accidents occur in less developed countries.

## Causes of fatal accidents



80+ raw causes were merged into 11 categories. Pedestrian-related accidents carry the highest fatal share (30.9%) despite forming only 3.1% of incidents. Surprisingly, Alcohol/Drugs sits below the overall fatal average at 5.4%.

## Temporal patterns

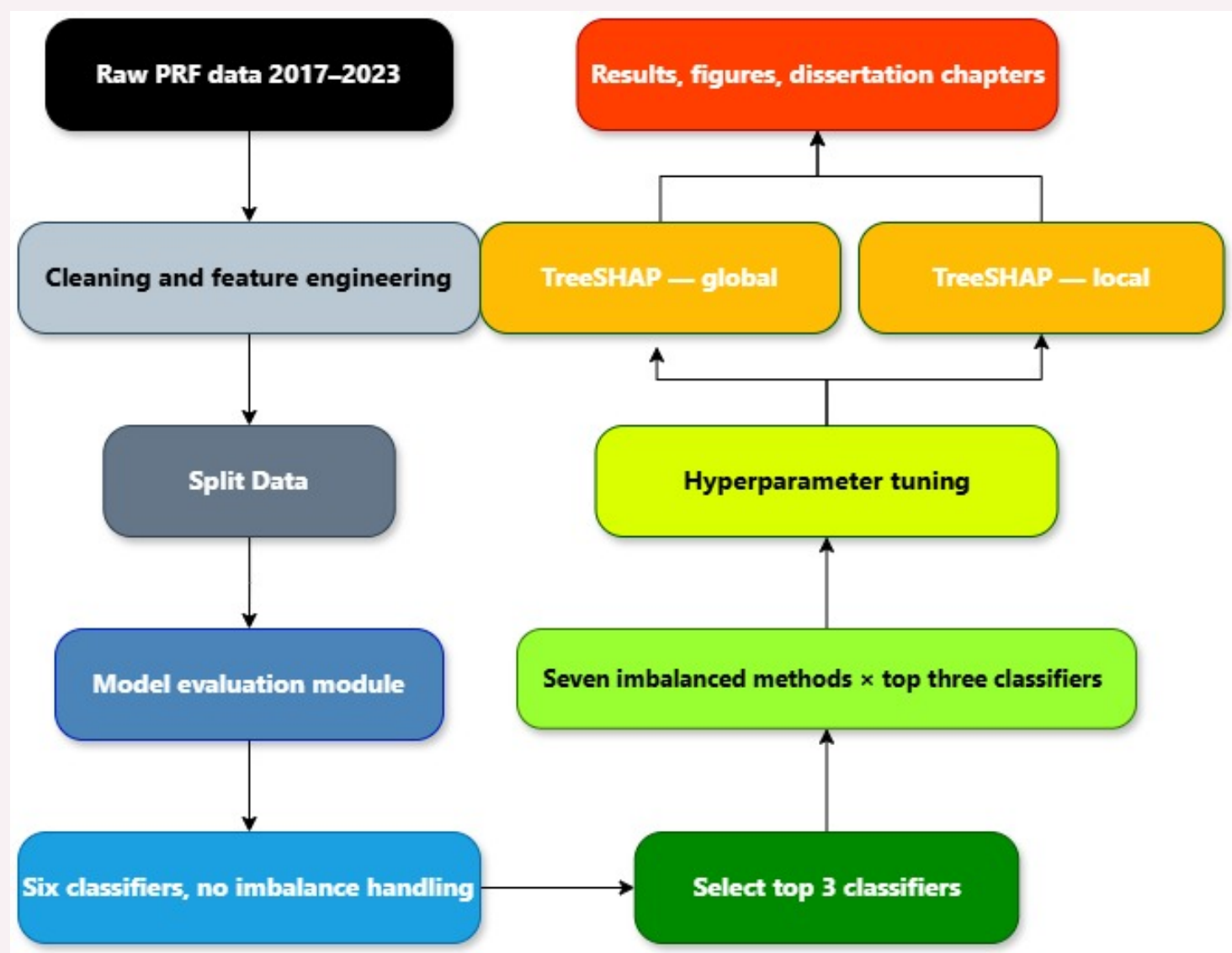


Weekend nights (22:00–05:00) show a clear spike, with Sundays consistently deadlier than Saturdays. This supports the use of a three-level **Day\_Type** feature (Weekday / Saturday / Sunday) rather than a single weekend class.

## Methodology

- 6 classifiers:** LR (baseline) + DT, RF, XGBoost, LightGBM, CatBoost
- 7 imbalance methods** applied to top 3 by macro-F1: none, random over/under-sampling, SMOTE, ADASYN, SMOTE+Tomek, class weighting
- Tuning:** Optuna with stratified cross-validation
- Metrics:** macro-F1, per-class F1, MCC, confusion matrices
- Explainability:** TreeSHAP — global feature ranking and local per-class explanations

## Artefact pipeline



## Phase timeline

- P1** (mid/late June) — finalise dataset, evaluation module
- P2** (late June) — baseline modelling, select top 3
- P3** (mid/late July) — 7 imbalance methods × top 3
- P4** (early August) — Optuna hyperparameter tuning
- P5** (early/mid August) — TreeSHAP analysis
- P6** (late Aug–early Sep) — writing and submission

## Expected contribution

Three contributions to PRF-based accident-severity modelling:

- The first multi-class severity model on the full national PRF dataset
- A systematic comparison of seven imbalance-handling methods across the best three classifiers.
- The first application of TreeSHAP to PRF national-level severity data.



Code & artefacts